Where the Matrix Links Come From: Group-Valued Links and Their Communicating Parts

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Abstract

The projected attention routes of the gauge-covariant transformer communicate through matrix links $L_{ij}^{(h)} = U_i O_h T_h U_j^{-1}$, which have rank at most d and are therefore not elements of $\mathrm{GL}(K)$. This note asks whether these links can be understood as parts of group-valued links $\Theta_{ij} \in \mathrm{GL}(K)$ that also contain a component that neither sends nor receives. A singular link cannot be a factor of a group element, so a literal product $\Theta = L \times N$ is impossible, but three exact constructions give the idea precise content. In the first, L is a Lie-algebra element and $\Theta_{ij} = U_i \exp(X_{ij}) U_j^{-1}$. The exponential of a rank-d algebra element splits exactly as the flat transport plus a rank-d channel term, $\Theta_{ij} = \Omega_{ij} + U_i O \varphi(TO) T U_j^{-1}$ with $\varphi(z) = (e^z - 1)/z$, so the non-communicating part is the identity on the kernel of the sending map and L is the first-order term. In the second, a primitive group link is compressed by frame-covariant projectors, and a block factorization relative to the parabolic subgroup that stabilizes the channel separates a communicating Levi block from a hidden Schur complement. In the third, the multi-head operator that the code already applies, $U_i O B_{ij} T U_j^{-1}$ with B_{ij} the head-diagonal routing weights, is itself generically group-valued, and its loops measure a cocycle defect of the attention logits. Under Option A the routing belongs to per-head source labels outside the exponent, and requiring that only sent content cross between agents makes the receiver carry the complement. The flat term then becomes a residual on the unsent directions only; with tiled heads, as the code configures them, there is no residual and Option A reparameterizes the current head sum. The function φ acts on weights rather than states, but the routing weights depend on the states, and a routed self-link yields a bounded block gate $e^{\beta_{ii}}$. For a channel of this form the linear link and the exponential link are two charts of the same embedded $\mathrm{GL}(d)$, related exactly by $O' = O\varphi(TO)$ wherever the principal logarithm exists, so the matrix link is the group-valued link in its affine chart and not a truncation of it; the charts differ at the boundary $\det = 0$, in components, and in optimization geometry. In one trained model the bare message operator sits at that boundary from early in training (stable rank 11–17 of 64 at the end, sign of the determinant not stable), while the flat-plus-channel link $I + M$ keeps all eigenvalues in the right half plane throughout. We rank the three constructions by how principled they are and by how closely they match standard attention, and we record what each predicts and leaves open. All identities were checked numerically in float64 at small dimension; no model was trained.

1 Setting and the question

Agent i carries a Gaussian belief $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ on $\mathbb{R}^K$ and an invertible frame U_i, with reference moments $\xi_i = U_i^{-1}\mu_i$ and $C_i = U_i^{-1}\Sigma_i U_i^{-\top}$. A passive frame change $g_i \in \mathrm{GL}(K)$ acts by $(\mu_i, \Sigma_i, U_i) \mapsto (g_i\mu_i, g_i\Sigma_i g_i^\top, g_i U_i)$ and fixes (ξ_i, C_i). The flat transport is $\Omega_{ij} = U_i U_j^{-1}$. Head h of a layer owns a sending map $T_h : \mathbb{R}^K \to \mathbb{R}^d$ and a receiving map $O_h : \mathbb{R}^d \to \mathbb{R}^K$, and the code requires $K = Hd$. The matrix link is

$$L_{ij}^{(h)} = U_i O_h T_h U_j^{-1}, \qquad L_{ij}^{(h)} \mapsto g_i L_{ij}^{(h)} g_j^{-1}, \tag{1}$$

and the projected predictive mean is $\mu_i^+ = r\mu_i + \sum_{h,j} \beta_{ij}^{(h)} L_{ij}^{(h)} \mu_j$ with $r \in \{0, 1\}$ set by the attention residual [1]. Stacking the heads gives square matrices $O = [O_1 \cdots O_H]$ and $T = [T_1; \ldots; T_H]$, both $K \times K$, and the head-diagonal routing matrix $B_{ij} = \bigoplus_h \beta_{ij}^{(h)} I_d$.

A group-valued link is a field $\Theta_{ij} \in \mathrm{GL}(K)$ with the same law $\Theta_{ij} \mapsto g_i \Theta_{ij} g_j^{-1}$, which is the lattice gauge field of Wilson [14]. The question is whether L_{ij} can be read as the part of some Θ_{ij} that takes part in sending and receiving, with a remainder that does not. To keep notation apart from the sending maps T_h, we write N for that remainder.

Two facts constrain any answer. First, $L_{ij}^{(h)}$ has rank at most $d < K$, so for any M the product $L^{(h)} M$ is singular, and no factorization $\Theta = L^{(h)} N$ of a group element exists. The relation between L and Θ must be additive, through a block of Θ, or infinitesimal, through the Lie algebra. Second, on causal support every source satisfies $j \le i$, so a closed directed walk can only be a self-loop, a point raised in the review of the attention manuscript [4]. Loop observables with more than one edge need reversed edges, which in lattice gauge theory are traversed with the inverse link [14, 11]. Inverses exist for group-valued links and not for singular ones, so a group-valued origin for L is also what gives the link field nontrivial loop observables on causal support.

2 Option A: the link is a Lie-algebra element

Definition 1 (Exponential link)**.** For an algebra field $X_{ij} \in \mathfrak{gl}(K)$ written in reference coordinates, set

$$\Theta_{ij} = U_i \exp(X_{ij}) U_j^{-1}. \tag{2}$$

The law $\Theta_{ij} \mapsto g_i \Theta_{ij} g_j^{-1}$ holds for every X_{ij}, because X_{ij} is expressed in reference coordinates and the frames carry all of the passive action. The identity $\det \exp(X) = e^{\operatorname{tr} X}$ makes Θ_{ij} invertible for every algebra element, singular ones included. The choice that reproduces the current links is the rank-d algebra element $X = O_h T_h$ for one head, or $X = \sum_h O_h T_h = OT$ for all heads with equal weight.

Lemma 2 (Exponential of a low-rank element)**.** *Let P be $K \times r$ and Q be $r \times K$, and let $\varphi(Z) = \sum_{n \ge 1} Z^{n-1}/n!$, which equals $(e^Z - I)Z^{-1}$ when Z is invertible. Then*

$$\exp(PQ) = I_K + P\varphi(QP)Q. \tag{3}$$

Proof. For $n \ge 1$, $(PQ)^n = P(QP)^{n-1}Q$ by regrouping the product. Summing the exponential series term by term gives $I_K + P\big(\sum_{n \ge 1} (QP)^{n-1}/n!\big)Q$, and the inner series converges for every $r \times r$ matrix. □

Proposition 3 (Flat transport plus a channel term)**.** *With $X_{ij} = O_h T_h$,*

$$\Theta_{ij} = \Omega_{ij} + \widetilde{L}_{ij}^{(h)}, \qquad \widetilde{L}_{ij}^{(h)} = U_i O_h \varphi(T_h O_h) T_h U_j^{-1}. \tag{4}$$

The following hold.

1. $\exp(O_h T_h)$ *acts as the identity on* $\ker T_h$*, so on the reference subspace that the head does not read, Θ_{ij} coincides with the flat transport Ω_{ij}.*
2. *The image of* $\exp(O_h T_h) - I$ *lies in* $\operatorname{im} O_h$*, so Θ_{ij} differs from Ω_{ij} only in the directions that the head writes.*

3. *Under a scaling* $X_{ij} = sO_hT_h$, $\Theta_{ij} = \Omega_{ij} + sL_{ij}^{(h)} + O(s^2)$, *so the matrix link of Eq.* (1) *is the first-order term.*

4. $\det \Theta_{ij} = \det(U_i U_j^{-1}) e^{\mathrm{tr}(T_h O_h)}$.

Proof. Equation (4) is Lemma 2 with $P = O_h$ and $Q = T_h$, conjugated by the frames. Items 1 and 2 are read off Eq. (3): the correction is $O_h(\cdots)T_h$. Item 3 follows from $\varphi(sZ) = I + O(s)$. Item 4 is $\det \exp(X) = e^{\mathrm{tr}\, X}$ with $\mathrm{tr}(O_hT_h) = \mathrm{tr}(T_hO_h)$. □

Proposition 3 gives the original question an exact answer. The remainder N of the question is the identity on the non-communicating subspace $\ker T_h$, and the communicating part is $\widetilde{L}_{ij}^{(h)}$. That part equals the matrix link to first order and carries the corrections $\varphi(T_hO_h)$ at higher order. Nothing in Θ_{ij} is hidden: every entry is fixed by the frames and by (O_h, T_h).

The two historical routes as orders of one link. Contracting Eq. (4) against routing weights gives the message

$$\sum_j \beta_{ij}\Theta_{ij}\mu_j = \underbrace{\sum_j \beta_{ij}\Omega_{ij}\mu_j}_{\text{flat transport}} + \underbrace{\sum_j \beta_{ij}L_{ij}\mu_j}_{\text{projected link}} + O(\|T_hO_h\|). \tag{5}$$

The first term is the transported message of the earlier transported route, and the second is the projected predictive message. Option A therefore places both routes inside one group-valued link, as its zeroth- and first-order terms. This is an observation about the form of the message, not a claim that either retired objective is recovered.

Routing inside the exponent. Setting $X_{ij} = OB_{ij}T = \sum_h \beta_{ij}^{(h)} O_hT_h$ instead dresses the algebra element with the routing weights, and Lemma 2 with $P = O$ and $Q = B_{ij}T$ gives

$$\exp(OB_{ij}T) = I + O\varphi(B_{ij}TO)B_{ij}T. \tag{6}$$

The first-order term is the multi-head operator of Section 4. Heads interact at second order through the cross-head blocks of TO. With routing inside the exponent, the weights no longer normalize a sum over sources, so the aggregation over j must be chosen separately. Section 5 resolves this by keeping the routing outside the exponent.

Loops. For a triangle with a reversed edge, the loop is $\Theta_{ij}\Theta_{jk}\Theta_{ik}^{-1} = U_i e^{X_{ij}} e^{X_{jk}} e^{-X_{ik}} U_i^{-1}$, and its conjugacy class, for example its trace, is frame invariant. When the algebra elements commute, which happens for $X = OBT$ with $TO = I$ since then $X_{ij} = OB_{ij}O^{-1}$, the loop equals

$$U_i O \exp\left(B_{ij} + B_{jk} - B_{ik}\right) O^{-1} U_i^{-1}, \tag{7}$$

so the curvature is the per-head additive defect $\beta_{ij}^{(h)} + \beta_{jk}^{(h)} - \beta_{ik}^{(h)}$. In general the loop follows the Baker-Campbell-Hausdorff series, and the commutators $[X_{ij}, X_{jk}]$ make it nonabelian.

2.1 Linear and exponential links are two charts of one group

The phrase "first-order term" in Proposition 3 invites the reading that the matrix link is a weak-field truncation of the group link. For a channel of this form the truncation loses nothing. Fix $O \in \mathbb{R}^{K\times d}$ and $T \in \mathbb{R}^{d\times K}$ with TO invertible, so that $\mathbb{R}^K = W \oplus W'$ with $W = \operatorname{im} O$ and $W' = \ker T$, and let $P = O(TO)^{-1}T$ be the projector onto W along W'. For $B \in \mathrm{GL}(d)$ define

$$\iota(B) = (I - P) + O\,B\,(TO)^{-1}T, \tag{8}$$

the operator that fixes W' pointwise and acts on W as B in the basis formed by the columns of O.

Proposition 4 (Two charts of one group)**.** *1. $\iota : \mathrm{GL}(d) \to \mathrm{GL}(K)$ is an injective homomorphism with $\det \iota(B) = \det B$. Its image $\mathcal{G}_W$ is the Levi factor of the stabilizer of the splitting $W \oplus W'$ that acts trivially on W'.*

2. Linear links. *For $C \in \mathbb{R}^{d\times d}$, $I + OCT = \iota(I_d + C\,TO)$, so $\det(I + OCT) = \det(I_d + C\,TO)$.*

3. Exponential links. $\exp(OT) = \iota(e^{TO})$.

4. Exact conversion. *Let $O' \in \mathbb{R}^{K\times d}$ be such that $I_d + TO'$ has no eigenvalue on the closed negative real axis $(-\infty, 0]$. Put $Z = \log(I_d + TO')$ (principal logarithm) and $O = O'\varphi(Z)^{-1}$. Then $\varphi(Z)$ is invertible, $TO = Z$, and*

$$\exp(OT) = I_K + O'T. \tag{9}$$

Proof. For item 1, $\iota(B)Oy = OBy$ because $(I - P)O = 0$ and $(TO)^{-1}TO = I$, and $\iota(B)w' = w'$ for $w' \in W'$ because $Pw' = 0$. An operator on $W \oplus W'$ is determined by its action on the two summands, so $\iota(B)\iota(B') = \iota(BB')$ and $\det \iota(B) = \det B \cdot 1$. For item 2, $(I + OCT)Oy = O(I + C\,TO)y$ and $(I + OCT)w' = w'$. For item 3, $(OT)^nO = O(TO)^n$ gives $\exp(OT)Oy = Oe^{TO}y$, and $\exp(OT)w' = w'$. For item 4, the eigenvalues of Z have imaginary part in $(-\pi, \pi)$, so $\varphi(Z)$, whose eigenvalues $(e^z - 1)/z$ vanish only at $z \in 2\pi i\mathbb{Z} \setminus \{0\}$, is invertible. Since $e^Z - I = TO' = Z\varphi(Z)$, we get $TO = TO'\varphi(Z)^{-1} = Z$, and Lemma 2 gives $\exp(OT) = I + O\varphi(TO)T = I + O'\varphi(Z)^{-1}\varphi(Z)T = I + O'T$. □

Item 4 says that the flat transport plus a linear channel term, $\Omega_{ij} + U_iO'TU_j^{-1}$, is exactly an exponential link for every O' in an open set that contains $O' = 0$. The two families are the same subgroup written in an affine chart ($B = I + C\,TO$) and a logarithmic chart ($B = e^Z$). The expansion $\Theta = \Omega + L + O(X^2)$ of Proposition 3 is the statement that the charts agree to first order at the identity, and with $\widetilde{L} = O\varphi(TO)T = \exp(OT) - I$ in reference coordinates the series $X = \log(I + \widetilde{L}) = \widetilde{L} - \frac{1}{2}\widetilde{L}^2 + \cdots$ converts one chart to the other. It is not a statement that the linear link approximates the group link. The same holds for the lattice relation $U = e^{aA}$, where the algebra-valued potential A is the logarithmic coordinate, with the difference that here the frames dress the potential, so that X is frame invariant in reference coordinates.

Corollary 5 (Tiled heads)**.** *If the stacked O and T are square and invertible, so that $W = \mathbb{R}^K$, the operator $M = O'T$ that the projected predictive route applies at fixed routing is itself a group element, $M \in \mathrm{GL}(K)$ when $\det M \neq 0$, and $M = \exp(X)$ for $X = OT$ exactly when M has a real logarithm, with $O = XT^{-1}$. The code's identity initialization $M = I$ is the origin $X = 0$ of the logarithmic chart. The product of r links along a path with a shared generator is $M^r = e^{rX}$, a one-parameter subgroup.*

The corollary uses $M^r = e^{rX}$ to restate the path-composition remark of the matrix-link construction: the r-edge path $U_{i_0}M^rU_{i_r}^{-1}$ is the group element of a one-parameter subgroup, not a separate object.

What the choice of chart does change. First, the affine chart is the whole matrix space. A step in (O', T) can carry $\det(I_K + O'T) = \det(I_d + TO')$ through zero (in the setting of Corollary 5, $\det M$), where the link becomes singular and the information carried by the W component is destroyed. The logarithmic chart cannot reach that boundary, because $\det e^Z = e^{\operatorname{tr} Z} > 0$. The same argument shows that the exponential chart covers only matrices with a real logarithm: it excludes negative-determinant components, which an optimizer in the affine chart reaches only by passing through $\det = 0$, and a real invertible matrix has a real logarithm only if its Jordan blocks for negative eigenvalues occur in pairs [5]. Second, the two charts give different optimization geometries, since the code's M-step takes Euclidean-type steps on (O', T), which is the affine chart. Third, they differ at order one for finite generators: at $X = I$, $e^X = eI$ while $I + X = 2I$. Fourth, these statements concern an unrouted operator with a single generator. The routed head sum $\sum_h \beta^{(h)} O_h T_h$ weights heads differently and does not lie in the one-generator family, and the hidden block and edge-dependent generators of Option B remain outside $\mathcal{G}_W$.

3 Option B: the link is a compression of a primitive group link

Here the group-valued link is taken as primary, $\Theta_{ij} = U_i G_{ij} U_j^{-1}$ with $G_{ij} \in \mathrm{GL}(K)$, and the channel maps act as covariant projectors. When $T_h O_h$ is invertible, define the reference projector $P_h = O_h (T_h O_h)^{-1} T_h$, which projects onto $\operatorname{im} O_h$ along $\ker T_h$, and its frame-dressed version $\Pi_{i,h} = U_i P_h U_i^{-1}$, which transforms by conjugation. Every link then decomposes into four covariant blocks,

$$\Theta_{ij} = \Pi_i \Theta_{ij} \Pi_j + \Pi_i \Theta_{ij} (I - \Pi_j) + (I - \Pi_i) \Theta_{ij} \Pi_j + (I - \Pi_i) \Theta_{ij} (I - \Pi_j), \tag{10}$$

where the first block is the communicating link, the last is the non-communicating remainder N of the question, and the middle two mix them. More generally, distinct receive and send projectors give the oblique compression $\Pi_i^R \Theta_{ij} \Pi_j^S$, and any rank-$d$ map whose range lies in $\operatorname{im} O_h$ and whose kernel contains $\ker T_h$ arises this way from some invertible G_{ij}.

Proposition 6 (Block factorization)**.** *In a reference basis adapted to $\mathbb{R}^K = W \oplus W'$ with $W = \operatorname{im} O_h$, write $G = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. If A is invertible, then*

$$G = \begin{pmatrix} I & 0 \\ CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} I & A^{-1}B \\ 0 & I \end{pmatrix}, \qquad S = D - CA^{-1}B, \tag{11}$$

and $\det G = \det A \det S$.

The middle factor lies in $\mathrm{GL}(d) \times \mathrm{GL}(K - d)$, the Levi factor of the parabolic subgroup that stabilizes W [10]. Its first block A is the communicating link, and S, the Schur complement [15], is the hidden part with the communicating directions eliminated. The outer unipotent factors carry the mixing. This is the precise multiplicative form of $\Theta = L \times N$: the product holds between the outer mixing factors and the block-diagonal pair (A, S), not between L and N directly.

Remark 7 (Reduction of structure group). Assigning the subspace $W_i = U_i \operatorname{im} O_h$ to every agent is a section of the associated Grassmannian bundle, and it reduces the structure group from $\mathrm{GL}(K)$ to the parabolic stabilizer. Reductions of this kind are how a Higgs field breaks a gauge symmetry. The analogy is formal; no potential or dynamics for W_i is proposed here.

What Option B needs and what it cannot see. The current model is the case $G_{ij} = G$ constant, with every head reading its own block. A genuine extension needs a model for G_{ij}: a learned relative-position element G_{i-j}, a state-dependent element $G(\xi_i, \xi_j)$, or a dynamical link with its own energy. Rotary embeddings are the closest precedent in standard practice, a relative-position orthogonal link inserted between queries and keys [12]. The code's `rope_on_value=True` also dresses the value link with the rotation, $V_i O_h T_h V_j^{-1} = U_i R_i^\top O_h T_h R_j U_j^{-1}$. The decomposition also says what cannot be learned. The blocks B, C and D never enter a message, so they receive no gradient from the routing or content path unless another term reads them. The projected gradient analysis gives one such term [2]. The ambient-alignment receiver energy pins the complement of each output range to zero, while the channel-likelihood energy leaves that complement at the prior. Under Option B, the identifiability of the hidden block is therefore a property of the chosen receiver energy, not of the link.

4 Option C: the multi-head operator is already group-valued

Proposition 8 (Head-sum link)**.** *The edge operator that the projected routes apply at fixed routing,*

$$\mathcal{C}_{ij} = \sum_h \beta_{ij}^{(h)} L_{ij}^{(h)} = U_i O B_{ij} T U_j^{-1}, \tag{12}$$

transforms as a link and lies in $\mathrm{GL}(K)$ *whenever the stacked* O *and* T *are invertible and every* $\beta_{ij}^{(h)} > 0$*. Its determinant is* $\det(U_i U_j^{-1}) \det(TO) \prod_h (\beta_{ij}^{(h)})^d$*.*

Proof. The identity $\sum_h \beta^{(h)} O_h T_h = OBT$ is block multiplication with $B = \bigoplus_h \beta^{(h)} I_d$, and the determinant is multiplicative. □

Invertibility of O and T is generic and holds at the code's identity initialization, where O_h and $T_h^\top$ are the coordinate block embeddings. Softmax weights are strictly positive. Proposition 8 is the remark in the attention manuscript that the multi-head operator can be full rank [3], made into a statement about group values. The heads are then the decomposition the question asks for: each head-diagonal block of $\mathcal{C}_{ij}$ is one head's link, the routing weights are the diagonal factor B_{ij}, and TO carries the mixing.

This is also exactly the value path of standard multi-head attention. With trivial frames, the multi-head output of a transformer is $\sum_h W_O^{(h)} \sum_j \beta_{ij}^{(h)} W_V^{(h)} x_j = \sum_j W_O B_{ij} W_V x_j$ [13], the frame-free case of Eq. (12) with $O = W_O$ and $T = W_V$.

Loops measure a cocycle defect of the logits. With $TO = I$ the head blocks commute, and the triangle loop with one reversed edge is

$$\mathcal{C}_{ij}\mathcal{C}_{jk}\mathcal{C}_{ik}^{-1} = U_i O \Big(\bigoplus_h \frac{\beta_{ij}^{(h)} \beta_{jk}^{(h)}}{\beta_{ik}^{(h)}} I_d \Big) O^{-1} U_i^{-1}. \tag{13}$$

For softmax routing $\beta_{ij} = e^{s_{ij}}/Z_i$, the logarithm of each head factor is

$$\log \frac{\beta_{ij}\beta_{jk}}{\beta_{ik}} = s_{ij} + s_{jk} - s_{ik} - \log Z_j, \tag{14}$$

because the receiver normalizer Z_i cancels between β_{ij} and β_{ik}. The loop is trivial exactly when the logits form a cocycle up to the intermediate normalizer. In the projected routes the routing is

$\beta_{ij}^{(h)} = \pi_{ij}^{(h)} \exp(-D_{ij}^{(h)}/\kappa_h)/Z_i^{(h)}$, with $D_{ij}^{(h)}$ the Gaussian KL between head h's query law of i and key law of j, so $s_{ij} = \log \pi_{ij} - D_{ij}/\kappa$ and

$$\log \frac{\beta_{ij}\beta_{jk}}{\beta_{ik}} = -\frac{D_{ij} + D_{jk} - D_{ik}}{\kappa} + \log \frac{\pi_{ij}\pi_{jk}}{\pi_{ik}} - \log Z_j. \tag{15}$$

The leading term is the triangle defect of the routing costs. A metric would make $D_{ij} + D_{jk} - D_{ik}$ nonnegative, but the Kullback-Leibler divergence obeys no triangle inequality, and here it compares a query law with a key law under different maps S_Q and S_K, so the defect can take either sign. With uniform priors, Eq. (15) says that the loop of the head-sum link measures how far the routing geometry is from additive along the path $i \to j \to k$, offset by the intermediate token's log normalizer. For dot-product logits $s_{ij} = q_i^\top k_j/\sqrt{d}$ the defect is instead a bilinear expression in the queries and keys of the three tokens. Reading attention weights as an abelian connection is an interpretation suggested by Eq. (13); Proposition 8 establishes only the algebra.

Limits. Group-valuedness in Option C is generic rather than structural. The condition number of $\mathcal{C}_{ij}$ is bounded by $\kappa(O)\,\kappa(B_{ij})\,\kappa(T)$, so it grows with the ratio of the largest to the smallest head weight on the edge and with the conditioning of the stacked maps, and concentrated routing pushes it toward the boundary of $\mathrm{GL}(K)$. A singular O or T, or $K \neq Hd$, removes group-valuedness altogether. In the one trained model measured in Section 6 the stacked maps are already ill conditioned, so the bare head-sum operator is invertible only nominally.

5 Aggregation, the residual, and the nonlinearity under Option A

This section settles how sources combine under Option A, shows when a residual follows from the construction and when it does not, and asks whether the function φ is related to the activations of a transformer MLP.

5.1 The complement split

Stack the heads of a layer into O of size $K \times Hd$ and T of size $Hd \times K$, allow $Hd \leq K$, and assume TO is invertible. Let $P = O(TO)^{-1}T$, the projector onto $W = \operatorname{im} O$ along $W' = \ker T$.

Lemma 9 (Complement split)**.**

$$\exp(OT) = (I - P) + O\,E\,T, \qquad E = e^{TO}(TO)^{-1}. \tag{16}$$

Proof. By Lemma 2, $\exp(OT) = I + O\varphi(TO)T = (I - P) + O\big[(TO)^{-1} + \varphi(TO)\big]T$, and $(TO)^{-1} + \varphi(TO) = (TO)^{-1} + (e^{TO} - I)(TO)^{-1} = e^{TO}(TO)^{-1}$ because every term is a function of TO. □

The first term is the non-communicating part of the question: it is the identity on W' and annihilates W. The second term reads the Hd sent coordinates $T\xi$, mixes them through the $Hd \times Hd$ matrix E, and writes them into W. Equation (16) is exact, so the group-valued link separates into a complement transport and a channel transport with no remainder.

5.2 Routing belongs outside the exponent

A link acts on one source state. A mixture over sources therefore needs the routing weights as probabilities over which state each part of the link carries, not as coefficients inside the generator. With $X_{ij} = OB_{ij}T$ in the exponent, the routed links $e^{X_{ij}}$ for different j act on different vectors and

no single group element aggregates them: a product $\prod_j e^{X_{ij}}$ has no source vector to act on, and a sum of exponentials is not normalized by $\sum_j \beta_{ij} = 1$. We therefore take the generator to be routing independent, $X = OT$ in reference coordinates, and attach the routing to source labels. The routed generator keeps its use as a loop observable (Eq. (7)) but does not define a message.

5.3 The aggregation rule

Following the predictive kernel of the projected predictive route [1], each head draws its own source label J^h with probability $\beta_{ij}^{(h)}$, independently across heads. In Eq. (16) each sent block $T_h\xi$ is read from its head's chosen source, and E mixes the sent blocks into W. The complement term needs a carrier of its own, a label J^0. The transported reference state is

$$\xi_i' = (I - P)\xi_{J^0} + \sum_h O\, E_{:,h}\, T_h \xi_{J^h}, \tag{17}$$

where $E_{:,h}$ is the block column of E that receives head h. The expectation over the independent labels is

$$\mathbb{E}[\xi_i'] = \mathbb{E}[(I - P)\xi_{J^0}] + \sum_h \sum_j \beta_{ij}^{(h)} O\, E_{:,h}\, T_h \xi_j. \tag{18}$$

Each sent block travels with its own head's label, so the cross-head mixing in E does not couple the head routings. Only the complement carrier remains to be chosen. The weights are functions of the states, $\beta_{ij}^{(h)} = \beta_{ij}^{(h)}(\mu, \Sigma)$, through the Gaussian comparison costs of queries and keys. Both projected routes evaluate them at the layer's input states and hold them fixed while the message is formed, which is exact inference in the declared cut model of the projected gradient analysis [2]. Equation (18) is therefore the expectation conditional on those states, and the state dependence of β does not alter the aggregation rule.

Proposition 10 (The communication principle fixes the complement carrier)**.** *If only sent content may cross between agents, the complement must be carried by the receiver, $J^0 = i$, and*

$$\mu_i^+ = U_i(I - P)U_i^{-1}\mu_i + \sum_h \sum_j \beta_{ij}^{(h)} U_i O\, E_{:,h}\, T_h U_j^{-1} \mu_j. \tag{19}$$

Proof. The complement $W' = \ker T$ holds exactly the directions that no head sends. A carrier $J^0 = j \neq i$ would place agent j's unsent coordinates in agent i's state, which contradicts the premise, so the receiver is the only admissible carrier. Equation (18) with $J^0 = i$, conjugated by the frames, gives the formula. □

The alternative, a complement carried by a routed label $J^0 \sim \beta^{(0)}$, gives $\sum_j \beta_{ij}^{(0)}(I - P)\xi_j$, which moves the unsent coordinates of other agents. That is a flat-transport message of the kind the earlier transported route used, and it is admissible only if agents are meant to read each other's full state.

5.4 Does the flat term produce a residual?

Proposition 10 answers the question. Under the communication principle the flat term becomes a residual, but only on the complement: the receiver keeps its own state exactly in the directions that no head writes and takes the transported message in the directions that heads do write. This projected residual $U_i(I - P)U_i^{-1}\mu_i$ differs from the full residual $r\mu_i$ that the code adds under `attention_residual`.

Corollary 11 (Tiled heads leave no residual)**.** *If the heads tile the fiber, $Hd = K$ with O and T invertible, then $P = I$, the complement is empty, and Eq. (19) reduces to*

$$\mu_i^+ = \sum_h \sum_j \beta_{ij}^{(h)} U_i O'_h T_h U_j^{-1} \mu_j, \qquad O'_h = O\, E_{:,h}, \tag{20}$$

the residual-free projected predictive head sum with receiving maps O'_h.

The code requires $K = Hd$, so with invertible stacked maps its heads tile the fiber, and Option A then implies no residual at all. Two consequences follow. First, the $r = 1$ residual of the code has no counterpart in Option A under the configuration the code uses, and the receiver's own state reaches μ_i^+ only through the self-weights $\beta_{ii}^{(h)}$. Second, in the tiled case Option A adds no expressive power to the predictive route: its head sum lies in the current function class, with receiving maps of the special form $O\,E_{:,h}$. We do not claim the converse, that every current set of receiving maps arises this way. Until the generator is made edge dependent, the content of Option A in the tiled case is the interpretation together with the loop observables of the group-valued link.

A residual of the projected kind requires channels that do not tile, $Hd < K$. Equation (19) then describes an architecture in which $K - Hd$ coordinates form a residual stream that no head writes and every head can read through T. In this reading the residual stream is the complement of the communication channels.

The same answer from Gaussian fusion. The projected gradient route reaches the same structure by a different argument [2]. Its channel-likelihood receiver keeps the prior in directions that no head observes, and in the tiled limit with a flat prior and equal receiving noise it recovers the residual-free predictive head sum. At finite precision, fusion also yields a partial residual in the observed directions. For an isotropic prior of precision α and message evidence of precision λ on W, the posterior mean is

$$\xi_i^+ = (I - P_W)\xi_i + P_W \frac{\alpha \xi_i + \lambda m_i}{\alpha + \lambda}, \tag{21}$$

with P_W the orthogonal projector onto W and m_i the message. The residual coefficient is the prior's share of precision, $\alpha/(\alpha + \lambda)$, in the observed directions and one in the unobserved directions, and Eq. (19) is the limit $\lambda/\alpha \to \infty$. Both derivations give a residual that is full on unobserved directions and partial or zero on observed ones. The additive residual of the code, $\mu_i + m_i$, is neither: it adds the full own state to the message in every direction without normalization. That form is the residual connection of standard transformers [8], whose justification is optimization rather than inference.

5.5 Is φ an activation function?

Not with the routing-independent generator $X = OT$. There φ acts on the $d \times d$ round-trip matrix $T_h O_h$, which is built from weights, the link is linear in the transported state, and in the tiled case φ is absorbed into the receiving maps (Corollary 11). The message is still nonlinear in the states, but only through $\beta(\mu, \Sigma)$, the way standard attention is nonlinear through its softmax.

The answer changes once the generator carries the routing. Because $\beta_{ij}^{(h)} = \beta_{ij}^{(h)}(\mu, \Sigma)$, the routed generator $X_{ij} = OB_{ij}T$ depends on the states, and $\varphi(B_{ij}TO)$ in Eq. (6) becomes a matrix function of them. On the self-link this is a well-defined single-source transformation, so the objection of the previous subsection to routing inside the exponent does not apply there. For tiled heads with $TO = I$,

$$\exp(OB_{ii}T) = \sum_h e^{\beta_{ii}^{(h)}(\mu,\Sigma)} O_h T_h, \tag{22}$$

so each head's block of the receiver state is multiplied by $e^{\beta_{ii}^{(h)}} \in (1, e)$, a gate whose argument is the self-routing weight $\beta_{ii}^{(h)} = \pi_{ii}^{(h)} e^{-D_{ii}^{(h)}/\kappa_h} / Z_i^{(h)}$. Here D_{ii} compares the token's own query and key laws, which differ because $S_Q \neq S_K$, and Z_i compares that cost with every competing source. This is a state-dependent nonlinearity produced by the group structure itself. It differs from a GELU in three ways. The gate reads the state only through the self-routing weight, which compares the token with its sources rather than reading the token's own coordinates. It acts on whole head blocks rather than coordinatewise. And it lies between 1 and e, so it can amplify a block by at most a factor of e but can never suppress it or change its sign.

A general state-dependent generator on the self-link, $\Theta_{ii} = U_i \exp(X(\xi_i)) U_i^{-1}$, does produce the gated form of modern activations. For a diagonal generator $X(\xi) = \mathrm{diag}(a(\xi))$ the update is $\xi_k \mapsto \xi_k e^{a_k(\xi)}$, a coordinatewise gate. Every positive gate has this form, since $x\sigma(x) = \exp(\log \sigma(x))x$, so the SiLU $x\sigma(x)$ [7] and the GELU $x\Phi(x)$ [9] are exponential self-links with generators $\log \sigma$ and $\log \Phi$. In lattice language the attention links are edge variables and a gated MLP is a state-dependent site term.

This is a rewriting, not a derivation. It places the MLP in the same group-valued language as the links but does not explain why the generator should be $\log \sigma$ or $\log \Phi$. A derivation needs a principle that fixes the generator. Template routing supplies one for a normalized gate, replacing independent two-way competitions by one softmax over Gaussian templates derived from a mixture prior [6].

6 A measurement in a trained model

Proposition 4 predicts which quantities decide whether the affine and logarithmic charts agree for a trained operator: the sign of $\det M$, the smallest singular value, and the position of the eigenvalues relative to the negative real axis. We evaluated them on one finished run of the projected predictive route, `54.73_wikitext-103_K64_depth-scaled` ($K = 64$, $H = 2$, $d = 32$, four layers, `relational_output_init="depth_scaled"`, attention residual on, seed 6, 15000 steps, test perplexity 54.73). The inputs are the final weights and the operator snapshots stored at twelve steps; the stacked operator is $M = \sum_h O_h T_h = OT$ in reference coordinates, which is the message operator when every head routes to one source, and $I + M$ is the flat-plus-channel link that the residual realizes. The scripts and outputs are in `link_origin_checks/run_54.73/`.

The bare operator sits at the singular boundary. The depth-scaled initialization gives $M = 2^{-3/2} I$, with all singular values equal to 0.354 and stable rank 64. By step 1000 the stable rank has fallen to 5–13, and from then on $\sigma_{\min}(M)$ lies between 10^{-5} and 4×10^{-3}, and at the final step $\sigma_{\max}$ is 1.2 to 1.9; the minimum over all later snapshots is 9.7×10^{-6}. At the final step, 90% of $\|M\|_F^2$ lies in 21–30 of the 64 singular directions, and the smallest singular values of the stacked O and T lie between 10^{-3} and 8×10^{-2}, so some directions are barely written and some barely read. The sign of $\det M$ is not stable: counted over consecutive snapshot pairs from step 1000 on, it changes 5, 5, 3 and 4 times in layers 0 to 3, 17 times in 40 pairs. At $\sigma_{\min} \sim 10^{-5}$ the sign is below the step size of the optimizer, so this shows that M sits at $\det = 0$ and not a clean sequence of crossings. At the final step the determinant is negative in layers 1–3, and since $\det(OBT) = \det O \det B \det T$ with $\det B > 0$, that sign holds for every positive routing weight. Between 4 and 20 eigenvalues of M have negative real part, mostly near zero, with the most negative at -0.15 to -0.38 at the end and -0.57 in layer 0 at step 1000.

The flat-plus-channel link stays inside the group. $I + M$ has positive determinant at every snapshot and layer, because every eigenvalue has positive real part, the smallest being 0.43 (layer 0, step 1000) and 0.54 or larger from step 1500 on. The principal logarithm therefore exists, and the hypothesis of item 4 of Proposition 4 holds throughout. The smallest singular value of $I + M$ over all later snapshots is 0.27, and at the final step the condition number is 4–6 in every layer (Table 1). Over the per-head weight box $(b_1, b_2) \in [0, 1]^2$, evaluated on a 41×41 grid at the final weights, $I + OBT$ never has $\det \leq 0$ and has $\sigma_{\min} \geq 0.38$ and condition number at most 6.1.

Layer	$\operatorname{sign}\det M$	$\sigma_{\min}(M)$	$\kappa(M)$	$\sigma_{\min}(I+M)$	$\min \operatorname{Re}\lambda(I+M)$
0	+	8.1×10^{-4}	2.0×10^{3}	0.38	0.62
1	−	3.0×10^{-5}	4.8×10^{4}	0.56	0.70
2	−	3.8×10^{-4}	3.2×10^{3}	0.52	0.74
3	−	1.7×10^{-3}	1.1×10^{3}	0.69	0.85

Table 1: Final-step chart quantities for $M = \sum_h O_h T_h$ and $I + M$ in the trained run.

Reading. In this run, the residual keeps the realized link in the identity component with a real logarithm, while the bare message operator does not. Two statements of the earlier sections change accordingly. Option C's remark that the head-sum operator is generically in $\mathrm{GL}(K)$ is nominally true here and numerically empty: the operator is invertible, but its condition number is 10^3–5×10^4 for equal head weights, its sign is set by a singular value of order 10^{-5}, and the stacked maps have $\kappa(O)$ between 126 and 2265 and $\kappa(T)$ between 19 and 172. And the group element that the model applies is $\Omega + L$, the form of Option A, not the bare L. The trained message operator also has a numerical null complement, which is the learned counterpart of the unwritten subspace W' of Section 5 with $K = Hd$, so the question of whether a projected residual stream with $Hd < K$ would train as well is no longer purely hypothetical. This observation does not show that the residual causes the well-conditioned $I + M$, since no ablation was run, and it does not show that a design with $Hd < K$ would perform better.

Limits. The measurement is one run, one seed, with $K = 64$ and two heads. Snapshots are 1000 to 1500 steps apart, so crossings between them are not seen. The routing-weight box was evaluated only at the final weights, and the realized per-edge routing was not used; the mean self weights at the end are 0.04–0.13, and at position 0 the only source is the token itself, so $B = I$ exactly and the unrouted operator is applied there. The archived earlier runs cannot be loaded by the current code, so there is no comparison run.

7 How the three options relate

The three constructions nest. Option C is Option B with the reference element $G_{ij} = OB_{ij}T$ and the head projectors. Option A with routing inside the exponent agrees with Option C at first order, $\exp(OB_{ij}T) = I + OB_{ij}T + O(\|B_{ij}\|^2)$, apart from the identity, which contributes the flat transport. Option B is the most general frame: it holds any group link together with any choice of channel projectors. Option A restricts the link to the exponential of the channel algebra element and so fixes the hidden block to the identity. Option C is the model as it stands, read as a link.

8 Ranking

Table 2 orders the options by how principled they are. Here the criteria are that the group property should be structural rather than generic, that every component should be determined by the model's own fields, that the construction should have a standard place in gauge theory, and that loop observables should be well defined. Table 3 orders the same options by closeness to the value path of standard multi-head attention.

Rank	Option	Reason
1	A, exponential link $U_i e^{X_{ij}} U_j^{-1}$	Group-valued for every algebra element. The non-communicating part is fixed to the flat transport, so nothing is hidden or unidentifiable. The matrix link is the first-order term, the transported and projected messages are its zeroth and first orders, and loops are defined through inverses. It mirrors the lattice expansion $U = e^{aA}$ [14, 11].
2	B, compression of a primitive link $\Pi_i \Theta_{ij} \Pi_j$	The standard gauge-theoretic picture: links primary, channels as a reduction of structure group, and an exact Levi factorization. It needs a model for G_{ij}, and its hidden block is invisible to the message unless the receiver energy reads it.
3	C, head-sum link $U_i O B_{ij} T U_j^{-1}$	Already computed by the code, and its loops have a clean reading. Group-valuedness is generic and degrades as routing concentrates, and nothing in the construction guarantees invertibility.

Table 2: Most to least principled.

Rank	Option	Relation to standard attention
1	C, head-sum link	Identical to the multi-head value path $\sum_j W_O B_{ij} W_V x_j$ when the frames are trivial. No new parameters.
2	A, exponential link	Standard attention at first order, plus a flat-transport value term $\sum_j \beta_{ij} \Omega_{ij} \mu_j$ that standard attention lacks, plus higher-order cross-head terms through TO.
3	B, compression of a primitive link	Requires a new link field G_{ij}. Its nearest standard relative, rotary position encoding, acts on scores rather than on values [12].

Table 3: Closest to least close to standard attention.

The two orders are close to reversed, and Option A sits in the middle of the second while leading the first. Read as a direction for building Θ_{ij}, this suggests Option A as the principled extension and Option C as the model's present content written in link form. Proposition 4 qualifies the first half. With a constant generator, Option A is not an extension of the unrouted operator but the same operator in logarithmic coordinates, so Option A adds content only through edge-dependent

generators, through chart-natural optimization, or through the boundary and component structure of the group that the affine chart ignores.

9 Open questions

Section 5 settles two questions that the first draft of this note left open. Aggregation under Option A attaches the routing to per-head source labels outside the exponent, and the communication principle makes the receiver the carrier of the complement, so the flat term is a residual on the complement only and vanishes when the heads tile the fiber. What remains open is whether a projected residual stream with $Hd < K$ trains as well as the current additive residual, and whether the self-gate of Eq. (22) or a general state-dependent self-link should replace part of the block MLP.

Identifiability under Option B depends on whether the chosen receiver energy reads the hidden block.

Proposition 4 turns the chart question into measurements, which Section 6 made on one run. It leaves open whether the same pattern, a bare message operator at the singular boundary and a flat-plus-channel link well inside the group, holds across seeds, scales, and head counts, and whether it persists without the attention residual, where the residual-free route would have to supply the identity on the null complement itself. Measuring the realized per-edge routing weights, and snapshots at finer spacing, would resolve the box evaluation and the crossing history that Section 6 could not.

The triangle defect of Eq. (15) is computable in trained models from logged routing costs and normalizers without new machinery, and it would show how close trained routing is to additive along paths.

10 Numerical checks

The script `link_origin_checks/checks.py`, with its output in `checks_output.txt`, evaluates each identity in float64 at $K = 6$, $H = 3$, $d = 2$ with random frames and maps. Lemma 2 holds to a relative error of 5.6×10^{-16}, its routed form Eq. (6) to 9.1×10^{-15}, the determinant identity to 1.5×10^{-16}, and the fixed action on $\ker T$ to 5.6×10^{-16}. The split of Eq. (4) holds to 7.3×10^{-16}, and the first-order defect $(\Theta - \Omega)/s - L$ scales linearly with s (2.1×10^{-2} at $s = 10^{-2}$, 2.2×10^{-3} at $s = 10^{-3}$). The block factorization holds to 2.6×10^{-11}, limited by the conditioning of the random block A, and its determinant identity to 4.3×10^{-15}. The loop formulas Eq. (13) and Eq. (7) hold to 2.7×10^{-16} and 7.4×10^{-16}, and the loop trace is unchanged under a frame change. The script `checks_aggregation.py`, with output in `checks_aggregation_output.txt`, checks Section 5. The complement split of Lemma 9 holds to 1.6×10^{-15} at $K = 7$ with a rank-4 channel, and $I - P$ fixes $\ker T$ and annihilates $\operatorname{im} O$ to 1.2×10^{-15}. Exact enumeration over the head-label tuple reproduces the closed form of Eq. (18) with a receiver-carried complement to 4.2×10^{-16}. With tiled heads $P = I$ to 6.1×10^{-16}, and the Option A message equals the head sum with receiving maps $O\,E_{:,h}$ exactly. The fusion residual of Eq. (21) holds to 5.6×10^{-16}, and the SiLU equals the exponential of its log-sigmoid generator to rounding. The script `checks_chart.py`, with output in `checks_chart_output.txt`, checks Section 2.1 at $K = 7$, $d = 3$ and, for the tiled case, $K = 6$. Item 4 of Proposition 4 holds to 5.2×10^{-16} and 2.5×10^{-15} in the relative error of Eq. (9) on the two random trials that satisfy the determinant condition, with $TO = Z$ to 4.4×10^{-15}. The embedding ι is a homomorphism to 2.1×10^{-16}, $\exp(OT) = \iota(e^{TO})$ and $I + OCT = \iota(I + C\,TO)$ hold to 5.3×10^{-16} and 3.1×10^{-16}, and $\det \iota(B) = \det B$ to 2.2×10^{-16}. In the tiled case $O' = e^X T^{-1}$ gives $O'T = e^X$ to 1.8×10^{-15}, and the identity initialization has $\log I = 0$. At $X = I$ the exponential

and affine charts differ by 36% in relative size. These checks ground the algebra at one small size; they say nothing about training behavior.

References

[1] R. C. Dennis, *The projected predictive route: theory and computational path*, technical note, manuscripts/Attention/projected_predictive.tex, 2026.

[2] R. C. Dennis, *The projected gradient route: receiver energies, entropy, and stable updates*, technical note, manuscripts/Attention/projected_gradient.tex, 2026.

[3] R. C. Dennis, *Gauge-covariant Gaussian attention through projected matrix links*, manuscript, manuscripts/Attention/GL(K)_attention_v2.tex, 2026.

[4] Referee panel report on the GL(K) attention v2 manuscript, manuscripts/reviews/2026-09-23-glk-v2/REVIEW.md, 2026.

[5] W. J. Culver, On the existence and uniqueness of the real logarithm of a matrix, *Proceedings of the American Mathematical Society* 17, 1146–1151, 1966.

[6] R. C. Dennis, *Template routing*, technical note, manuscripts/Attention/template_routing.tex, 2026.

[7] S. Elfwing, E. Uchibe, and K. Doya, Sigmoid-weighted linear units for neural network function approximation in reinforcement learning, *Neural Networks* 107, 3–11, 2018.

[8] K. He, X. Zhang, S. Ren, and J. Sun, Deep residual learning for image recognition, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778, 2016.

[9] D. Hendrycks and K. Gimpel, Gaussian error linear units (GELUs), arXiv:1606.08415, 2016.

[10] J. E. Humphreys, *Linear Algebraic Groups*, Graduate Texts in Mathematics 21, Springer, 1975.

[11] J. B. Kogut, An introduction to lattice gauge theory and spin systems, *Reviews of Modern Physics* 51, 659–713, 1979.

[12] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu, RoFormer: enhanced transformer with rotary position embedding, arXiv:2104.09864, 2021.

[13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, Attention is all you need, *Advances in Neural Information Processing Systems* 30, 2017.

[14] K. G. Wilson, Confinement of quarks, *Physical Review D* 10, 2445–2459, 1974.

[15] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.